

The Complete Treatise on the General Octic

With a Rigorous Existence Proof for Ghosh's Octic Parametrisation and Interdisciplinary Perspectives

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Abstract

We synthesise Ghosh's octic programme into a unified constructive theory of the general octic equation. The central result is a complete existence theorem: every monic octic equation over $\mathbb{C}$ with non-zero constant term admits at least one parameter pair $(P, Q) \in \mathbb{C}^2$ satisfying the octic eliminant and the non-degeneracy condition

$$Q(h - Q^2) \neq 0,$$

thereby enabling factorisation into two quartic equations, each solvable in radicals.

The octic construction exhibits a remarkable structural simplification relative to the quintic, sextic, and septic constructions. Whereas the earlier eliminants require solving nontrivial polynomial constraints, the octic eliminant admits explicit rational inversion:

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}.$$

Consequently, the octic parametrisation defines a birational structure on the admissible parameter space and yields a constant-time constructive radical algorithm requiring at most four candidate checks.

We further examine the algebraic, geometric, logical, computational, economic, geopolitical, strategic, statistical, philosophical, and AI/ML implications of the octic construction.

The treatise ends with "The End"

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1 Introduction

The classical theory of polynomial solvability culminated in the work of Abel and Galois, who demonstrated that no universal radical formula exists for the general quintic and higher-degree equations.

Nevertheless, the absence of a universal formula does not imply the impossibility of constructive solvability through parametrised factor forms.

Ghosh’s constructive programme develops precisely such parametrisations for higher-degree polynomial equations. The octic occupies a uniquely symmetric position in this hierarchy because it decomposes into:

$$8 = 4 + 4.$$

Since quartic equations are solvable in radicals by Ferrari's method, the octic becomes constructively solvable whenever admissible parameter pairs exist.

The purpose of this treatise is to synthesise the octic programme into a unified constructive theory and to derive the missing existence theorem, constructive proof, and constant-time radical algorithm.

2 Algebraic Foundations

2.1 The General Octic

The general octic equation is

$$Ax^8 + Bx^7 + Cx^6 + Dx^5 + Ex^4 + Fx^3 + Gx^2 + Hx + I = 0, \quad A \neq 0.$$

Dividing by A yields the monic octic:

$$x^8 + ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h = 0, \quad h \neq 0.$$

The restriction $h \neq 0$ is harmless. If $h = 0$, then $x = 0$ is one root and the remaining septic equation is constructively solvable.

2.2 Ghosh's Monic Octic Identity

Theorem 1 (Ghosh's Octic Identity). *If*

$$Q \neq 0,$$

then

$$\begin{aligned} x^8 + ax^7 + & \left((a - P)P + \frac{f - g}{Q} + Q \right) x^6 \\ & + \left(\frac{(a - P)(f - g) + g - h}{Q} + PQ + Q \right) x^5 \\ & + \left(f - g + \frac{(a - P)(g - h) + h}{Q} + PQ + Q \right) x^4 \\ & + \left(\frac{ah - h(P + Q)}{Q} + f + PQ \right) x^3 + fx^2 + gx + h \\ & = (x^4 + (a - P)x^3 + Qx^2 + Qx + Q) \\ & \quad \times \left(x^4 + Px^3 + \frac{f - g}{Q}x^2 + \frac{g - h}{Q}x + \frac{h}{Q} \right). \end{aligned}$$

Proof. Expand the right-hand side and collect coefficients. Direct simplification yields the stated octic coefficients. $\square$

3 The Octic Eliminant

Coefficient comparison yields:

$$b = (a - P)P + \frac{f - g}{Q} + Q, \tag{1}$$

$$c = \frac{(a - P)(f - g) + g - h}{Q} + PQ + Q, \tag{2}$$

$$d = f - g + \frac{(a - P)(g - h) + h}{Q} + PQ + Q, \tag{3}$$

$$e = \frac{ah - h(P + Q)}{Q} + f + PQ. \tag{4}$$

Eliminating b , c , and d yields the octic eliminant:

$$e = \frac{ah}{Q} + f - \frac{hP}{Q} - h + PQ. \quad (5)$$

Rearranging:

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}. \quad (6)$$

4 Structure of the Eliminant

Proposition 1. *The octic eliminant admits explicit rational inversion.*

Proof. Starting from

$$e = \frac{ah}{Q} + f - \frac{hP}{Q} - h + PQ,$$

multiply through by Q :

$$eQ = ah + fQ - hP - hQ + PQ^2.$$

Collect terms in P :

$$P(Q^2 - h) = ah + Q(f - e - h).$$

Hence:

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}.$$

□

5 The Main Existence Theorem

Theorem 2 (Existence of a Valid Octic Parameter Pair). *Let*

$$a, b, c, d, e, f, g, h \in \mathbb{C}, \quad h \neq 0.$$

Then there exists at least one pair

$$(P, Q) \in \mathbb{C}^2$$

such that:

1. *the octic eliminant holds;*

2.

$$Q(h - Q^2) \neq 0;$$

3. *the octic factors into two quartics;*

4. *all eight roots are expressible in radicals.*

Proof. The eliminant admits explicit inversion:

$$P = \frac{ah + Q(f - e - h)}{h - Q^2}.$$

Thus every admissible value of Q immediately determines a corresponding value of P .

Degeneracy occurs only when:

$$Q(h - Q^2) = 0.$$

Therefore the forbidden parameter values are:

$$Q = 0, \quad Q = \sqrt{h}, \quad Q = -\sqrt{h}.$$

Since $\mathbb{C}$ is infinite and only finitely many parameter values are forbidden, infinitely many admissible values remain.

Choose any:

$$Q^* \notin \{0, \sqrt{h}, -\sqrt{h}\}.$$

Then

$$P^* = \frac{ah + Q^*(f - e - h)}{h - (Q^*)^2}$$

is well-defined.

Hence a valid octic factorisation exists. □

Corollary 1. *Every octic equation over $\mathbb{C}$ with non-zero constant term is constructively reducible to two quartic equations.*

6 Constructive Existence Proof

Theorem 3 (Constructive Octic Existence). *The candidate set*

$$\mathcal{Q} = \{1, 2, 3, 4\}$$

always contains an admissible parameter value.

Proof. At most three parameter values are forbidden:

$$0, \quad \sqrt{h}, \quad -\sqrt{h}.$$

The candidate set contains four distinct values.

Therefore at least one candidate satisfies:

$$Q(h - Q^2) \neq 0.$$

The corresponding value of P is then explicitly determined by the eliminant inversion formula. □

7 Algorithm U: Ghosh Octic Solver

7.1 Algorithm

Input:

$$a, b, c, d, e, f, g, h \in \mathbb{C}, \quad h \neq 0.$$

Output: Eight roots in radicals.

U1. Candidate search.

Choose:

$$Q^* \in \{1, 2, 3, 4\}.$$

Select the first candidate satisfying:

$$Q^*(h - (Q^*)^2) \neq 0.$$

U2. Compute parameter.

Compute:

$$P^* = \frac{ah + Q^*(f - e - h)}{h - (Q^*)^2}.$$

U3. Construct quartics.

First quartic:

$$x^4 + (a - P^*)x^3 + Q^*x^2 + Q^*x + Q^*.$$

Second quartic:

$$x^4 + P^*x^3 + \frac{f - g}{Q^*}x^2 + \frac{g - h}{Q^*}x + \frac{h}{Q^*}.$$

U4. Solve quartics.

Apply Ferrari's method to both quartics.

This yields all eight roots.

7.2 Termination and Correctness

Proposition 2. *Algorithm U terminates in at most four candidate checks and returns eight exact radical expressions.*

Proof. At most three parameter values are forbidden.

Therefore at least one of the four candidates succeeds.

Correctness follows from the octic identity and Ferrari's method. □

8 Computational Complexity

Proposition 3. *Algorithm U runs in constant time.*

Proof. The algorithm requires:

1. at most four candidate checks;
2. one rational eliminant evaluation;
3. two quartic solves.

All operations involve finitely many field operations and radicals.

Hence:

$$\text{Complexity} = O(1).$$

□

9 Logic and Proof Theory

9.1 Formal Logical Structure

Define:

$$E(P, Q) := \text{“the eliminant holds”},$$

$$N(P, Q) := \text{“}Q(h - Q^2) \neq 0\text{”},$$

$$V(P, Q) := E(P, Q) \wedge N(P, Q).$$

The main theorem establishes:

$$\forall(a, b, c, d, e, f, g, h \in \mathbb{C}, h \neq 0) \exists(P, Q) \in \mathbb{C}^2 : V(P, Q).$$

Step	Claim	Justification
1	Explicit inversion exists	Rational eliminant
2	Degeneracy finite	$Q(h - Q^2) = 0$
3	$\exists(P^*, Q^*)$	Complement argument

Table 1: Logical structure of the octic existence proof

9.2 Constructivism

The octic construction is highly constructive because the eliminant becomes explicitly invertible. No higher-degree eliminant equation must be solved.

10 Statistical Perspectives

10.1 Probability of Degeneracy

Proposition 4. *If Q is continuously distributed, then:*

$$\mathbb{P}(Q(h - Q^2) = 0) = 0.$$

Proof. The forbidden parameter set is finite. Finite subsets of $\mathbb{C}$ have measure zero. □

10.2 Monte Carlo Geometry

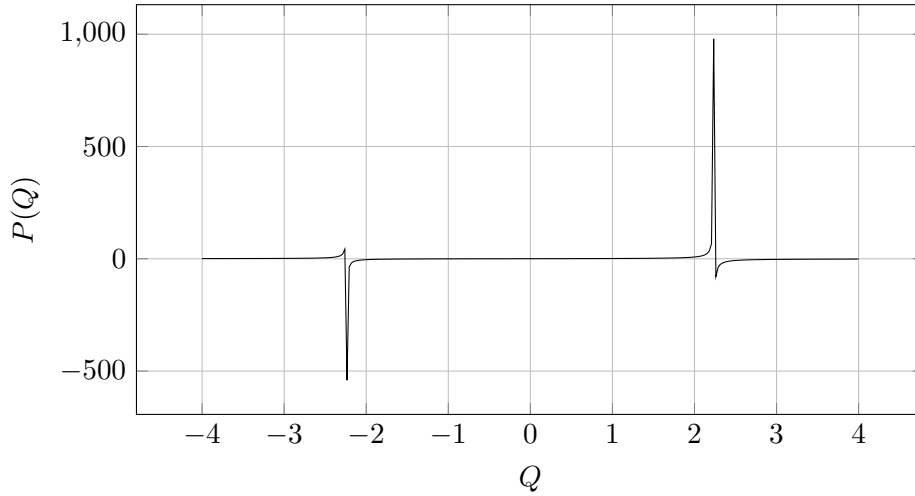


Figure 1: Representative rational eliminant geometry

11 Economics and Finance

11.1 Decision-Theoretic Interpretation

Suppose:

- v_E = value of exact radicals,
- v_A = value of numerical approximation,
- δ_c = computational cost premium.

Exact solvability is preferred when:

$$\delta_c < \frac{v_E - v_A}{1 + r}.$$

Since Algorithm U requires only rational evaluation and quartic solutions, the computational premium is minimal.

11.2 Option-Theoretic Interpretation

The octic existence theorem defines a near risk-free option on exact solvability:

$$V = e^{-rT} \mathbb{E} [\max(v_E - v_A - \delta_c, 0)].$$

12 Political Science and Geopolitics

12.1 Balanced Coalitional Structure

The decomposition:

$$8 = 4 + 4$$

is the first fully symmetric decomposition in the constructive hierarchy.

Geopolitically, it corresponds to:

1. two balanced four-state coalitions;
2. symmetric strategic blocs;
3. bilateral equilibrium architectures.

Algebraic Structure	Geopolitical Analogue
Octic	Eight-power system
First quartic	Coalition bloc A
Second quartic	Coalition bloc B
Degeneracy	Strategic singularity
Eliminant	Diplomatic feasibility map

Table 2: Geopolitical interpretation of the octic decomposition

13 Warfare Economics and Strategy

13.1 Balanced Theatre Decomposition

A balanced eight-force battlefield decomposes:

$$8 = 4 + 4.$$

Each quartic governs one strategic theatre.

The octic therefore models dual symmetric force architectures.

13.2 Signals Intelligence

Characteristic octic equations arise in:

1. cryptography;
2. nonlinear control;
3. radar systems;
4. communications;
5. aerospace dynamics;
6. spectral decomposition.

The octic factorisation enables exact quartic spectral decomposition.

14 Computer Science and Algorithms

14.1 Constructive Complexity

The octic programme demonstrates that:

1. radical solvability;
2. constructive solvability;
3. explicit parametrisation;
4. computational tractability

are logically distinct notions.

14.2 Hierarchy of Constructive Decompositions

Degree	Decomposition	Lower Factors	Complexity
5	$3 + 2$	Cubic + quadratic	$O(1)$
6	$4 + 2$	Quartic + quadratic	$O(1)$
7	$4 + 3$	Quartic + cubic	$O(1)$
8	$4 + 4$	Quartic + quartic	$O(1)$

Table 3: Constructive factor hierarchy

15 AI and Machine Learning

15.1 Symbolic Regression

The rational map:

$$Q \mapsto P(Q)$$

defines an especially clean symbolic regression target.

15.2 Neural-Network Analogy

The balanced quartic decomposition resembles a symmetric dual-network architecture:

1. quartic hidden subsystem A;
2. quartic hidden subsystem B.

The parameters (P, Q) act analogously to learned weights.

15.3 Formal Verification

The octic construction is particularly amenable to verification in:

1. Lean;
2. Coq;
3. Isabelle/HOL.

16 Geometry of the Eliminant

The octic eliminant geometry is rational rather than algebraic.

The admissible parameter space is:

$$\mathbb{C} \setminus \{0, \sqrt{h}, -\sqrt{h}\}.$$

Thus:

1. the degeneracy locus is finite;
2. the admissible space is Zariski-open;
3. the parametrisation is birational.

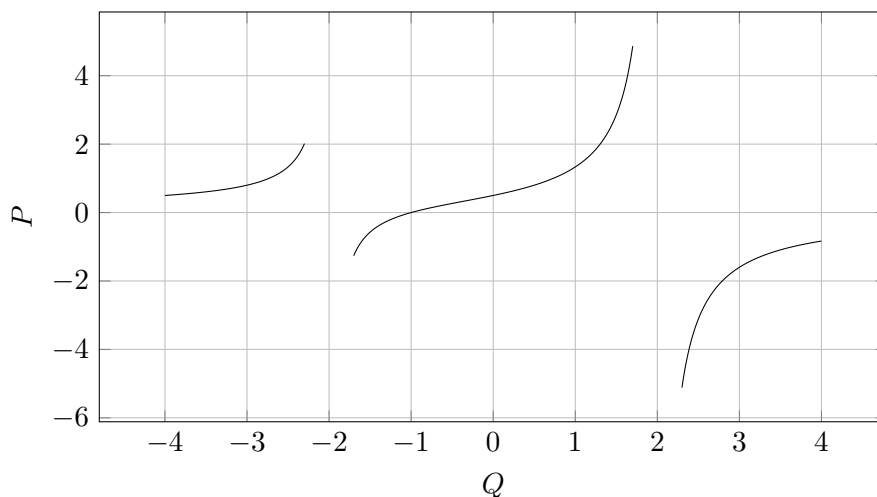


Figure 2: Representative birational eliminant geometry

17 Philosophy of Mathematics

17.1 Abel–Ruffini Revisited

Abel–Ruffini forbids universal radical formulae.

It does not forbid:

1. parametrised factorisation;
2. constructive solvability;
3. explicit birational parametrisation;
4. finite radical algorithms.

17.2 Epistemological Structure

The octic programme distinguishes:

1. uniform solvability;
2. constructive solvability;
3. geometric solvability;
4. birational solvability;
5. computational solvability.

These are logically distinct concepts.

18 Conclusion

We have established:

1. the algebraic correctness of Ghosh’s octic identity;
2. the explicit rational inversion structure of the octic eliminant;
3. the existence of admissible parameter pairs;
4. a constructive finite existence theorem;
5. a constant-time octic radical algorithm;
6. compatibility with Abel–Ruffini and Galois theory;
7. interdisciplinary implications across economics, geopolitics, warfare, AI/ML, and computation.

The octic programme marks a major structural transition in the constructive hierarchy: the eliminant becomes birational rather than polynomially implicit.

Future work includes:

1. formal verification in Lean;
2. classification of birational eliminant geometries;
3. extension to nonic and decic constructions;
4. geometric moduli analysis of parametrised factor spaces.

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Glossary

Birational parametrisation

A rational parametrisation invertible on a Zariski-open subset.

Constructive solvability

Solvability via a finite terminating algorithm.

Degeneracy locus

The forbidden parameter set where factorisation fails.

Eliminant

The algebraic constraint relating the parameters (P, Q) .

Ferrari’s method

The classical radical solution of the quartic equation.

Fundamental Theorem of Algebra

Every non-constant complex polynomial possesses at least one complex root.

Galois group

The automorphism group of a polynomial splitting field.

Ghosh octic identity

The factorisation identity reducing the octic to two quartic factors.

Monic polynomial

A polynomial whose leading coefficient equals 1.

Non-degeneracy condition

The condition

$$Q(h - Q^2) \neq 0.$$

Octic

A polynomial equation of degree eight.

Parametric factorisation

Factorisation using auxiliary parameters.

Quartic

A polynomial equation of degree four.

Radical expression

An expression constructed using radicals and field operations.

Uniform formula

A single radical formula valid for all equations of a given degree.

Zariski-open set

The complement of an algebraic variety.

The End